

Lecture 9: Density Estimation, and Classification

ML/AI for Engineering Applications
ENV 6932

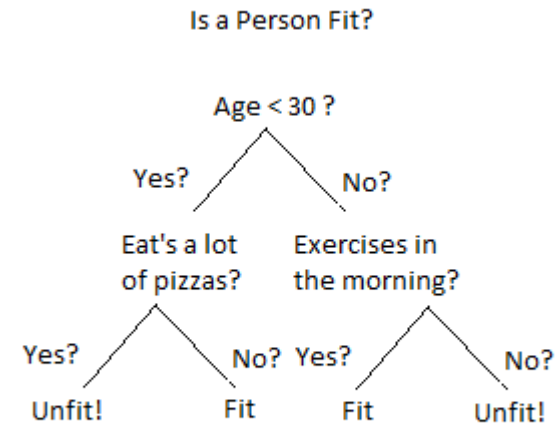
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Classification

- Recall
- Decision Tree

Decision Tree Algorithm

- Decision Trees (DTs) are a non-parametric supervised learning method.
- Used for classification and regression.



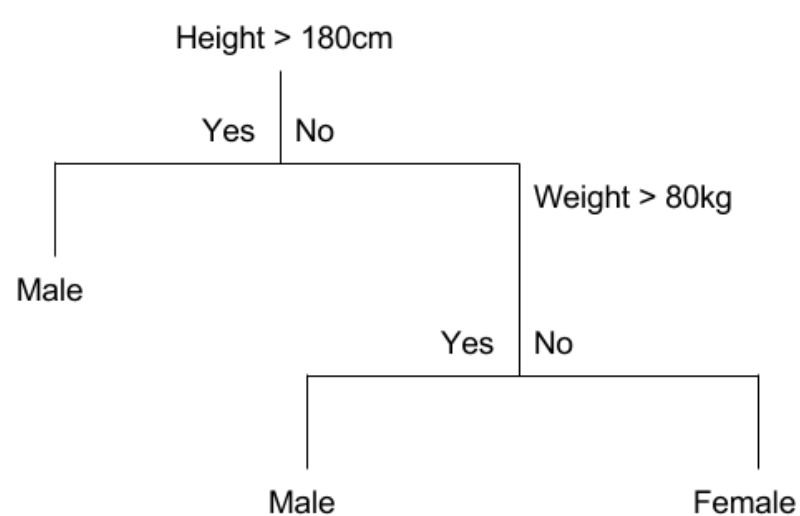
Source: xoriant.com

Elements of Decision Tree

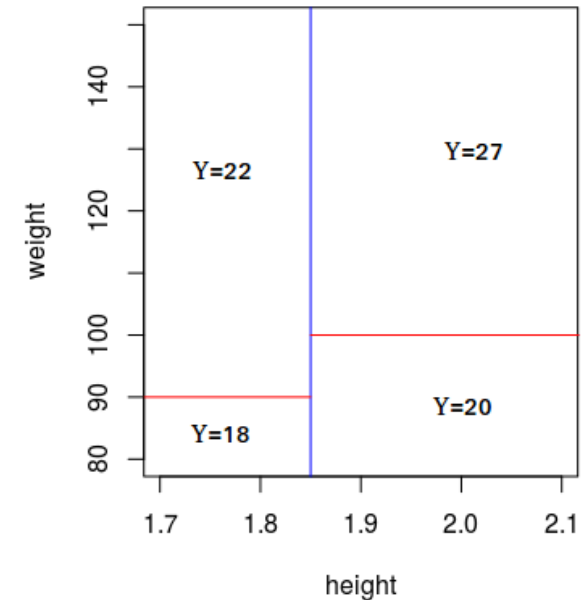
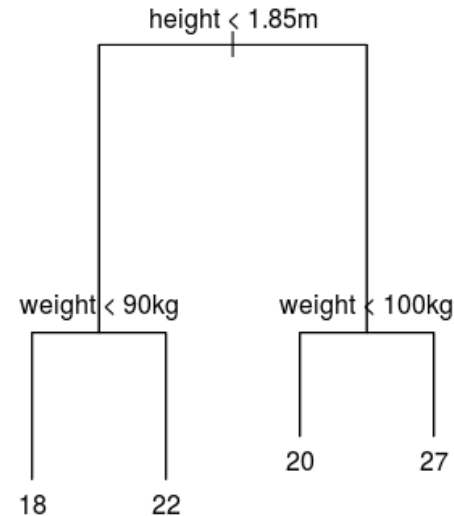
- **Decision Tree consists of:**
- **Nodes:** Test for the value of a certain attribute.
- **Edges/ Branch:** Correspond to the outcome of a test and connect to the next node or leaf.
- **Leaf nodes:** Terminal nodes that predict the outcome (represent class labels or class distribution).

Two Main Types of Decision Trees

- **Classification trees** (Yes/No types)
- **Regression trees** (Continuous data types)



Example of a Classification Tree



Example of Regression Tree

- Classification And Regression Trees or CART

Decision Tree Learning Algorithm

- Select an attribute from the dataset
- Calculate the significance of attribute in splitting of data
- Split data based on the value of the best attribute
- Go to Step 1

Ways to Split a Decision Tree

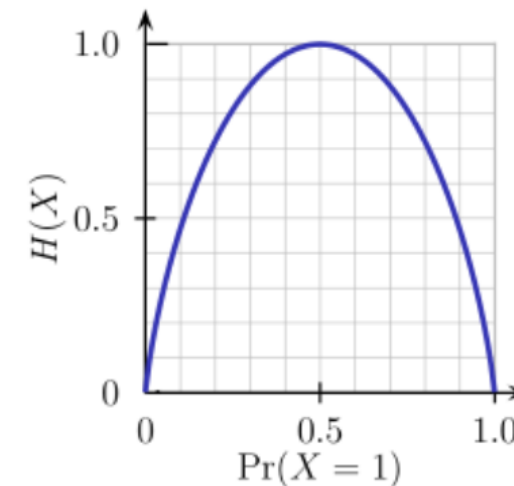
- Continuous Target Variable
 - Reduction in Variance
- Categorical Target Variable
 - Entropy (a measure of randomness)
 - Gini Impurity
 - Information Gain
 - Chi-Square

The Concept of Entropy

- Entropy: the measure of disorder

$$E(S) = \sum_{i=1}^c -p_i \log_2 p_i$$

- p_i : is simply the frequentist probability of an element/class 'i' in our data.
- **Simple example:** we have two classes , a positive class and a negative class. 'i' could be either (+) or (-).
- Suppose we have a total of 100 data points in our dataset with 30 belonging to the positive class and 70 belonging to the negative class:
 - $P(+)= 3/10$
 - $P(-) =7/10$



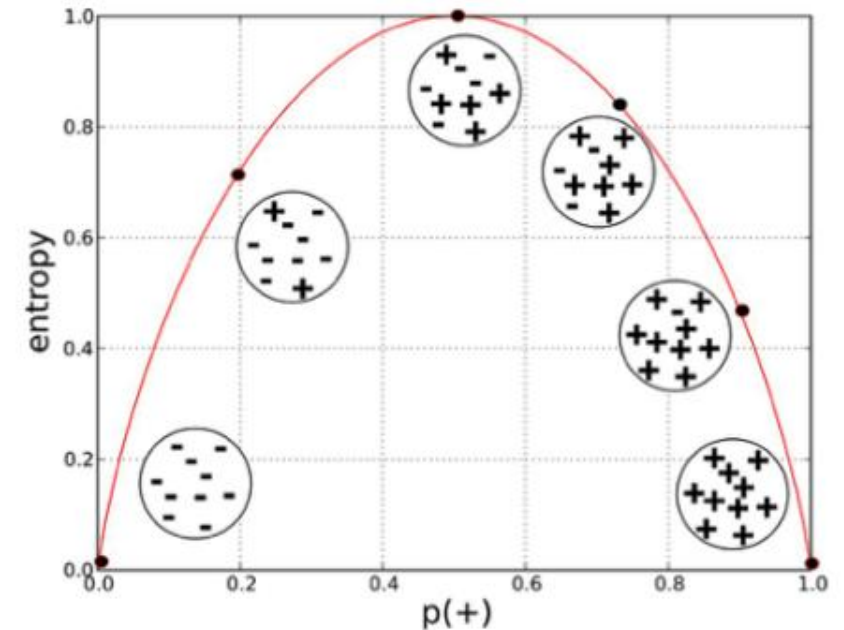
<https://www.kdnuggets.com/2020/01/decision-tree-algorithm-explained.html>

Example of Entropy

- Entropy:

$$-\frac{3}{10} \times \log_2\left(\frac{3}{10}\right) - \frac{7}{10} \times \log_2\left(\frac{7}{10}\right) \approx 0.88$$

- 0.88 is a high entropy , a high level of disorder.
- Entropy is lowest at the extremes (no + or only positive instances)
- Entropy is highest in the middle (bubble is evenly split between positive and negative instances, extreme disorder , no majority).



The x-axis represents the proportion of data points belonging to the positive class, y-axis axis measures their respective entropies.

Information Gain

- How to measure the reduction in disorder?
- A metric to measure the reduction of disorder in our target variable/class given additional information (features/independent variables) about it.
- This is called Information Gain:

$$IG(Y, X) = E(Y) - E(Y|X)$$

Information Gain from X on Y

- Decision trees use entropy and information gain to decide on what features to split their nodes on.

Example: Liability based on Credit Score

- Target variable: Liability has two values “Normal” and “High”
- Features: only one feature, Credit Rating with three values (“Excellent”, “Good” and “Poor”).

Credit Rating		Liability		
		Normal	High	Total
Excellent	3	1	4	
Good	4	2	6	
Poor	0	4	4	
Total	7	7	14	

Example: Liability based on Credit Score

- Target variable: Liability has two values “Normal” and “High”
- Features: only one feature, Credit Rating with three values (“Excellent”, “Good” and “Poor”).

$$\begin{aligned} E(\text{Liability}) &= -\frac{7}{14}\log_2\left(\frac{7}{14}\right) - \frac{7}{14}\log_2\left(\frac{7}{14}\right) \\ &= -\frac{1}{2}\log_2\left(\frac{1}{2}\right) - \frac{1}{2}\log_2\left(\frac{1}{2}\right) \\ &= 1 \end{aligned}$$

Credit Rating		Liability		
		Normal	High	Total
Excellent	3	1	4	
Good	4	2	6	
Poor	0	4	4	
Total	7	7	14	

Example: Liability based on Credit Score

- Calculate the entropy of the target variable Liability given additional information about credit score.

Credit Rating		Liability		
		Normal	High	Total
Excellent	3	1	4	
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Poor	0	4	4	
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Example: Liability based on Credit Score

- Calculate the entropy of the target variable Liability given additional information about credit score.

$$E(\text{Liability} \mid CR = \text{Excellent}) = -\frac{3}{4}\log_2\left(\frac{3}{4}\right) - \frac{1}{4}\log_2\left(\frac{1}{4}\right) \approx 0.811$$

$$E(\text{Liability} \mid CR = \text{Good}) = -\frac{4}{6}\log_2\left(\frac{4}{6}\right) - \frac{2}{6}\log_2\left(\frac{2}{6}\right) \approx 0.918$$

$$E(\text{Liability} \mid CR = \text{Poor}) = -0\log_2(0) - \frac{4}{4}\log_2\left(\frac{4}{4}\right) = 0$$

Weighted Average:

$$E(\text{Liability} \mid CR) = \frac{4}{14} \times 0.811 + \frac{6}{14} \times 0.918 + \frac{4}{14} \times 0$$

$$= 0.625$$

Credit Rating	Liability		
	Normal	High	Total
Excellent	3	1	4
Good	4	2	6
Poor	0	4	4
Total	7	7	14

Example: Liability based on Credit Score

- Information Gain:

$$IG(\textit{Liability}, CR) = E(\textit{Liability}) - E(\textit{Liability} \mid CR)$$

$$= 1 - 0.625$$

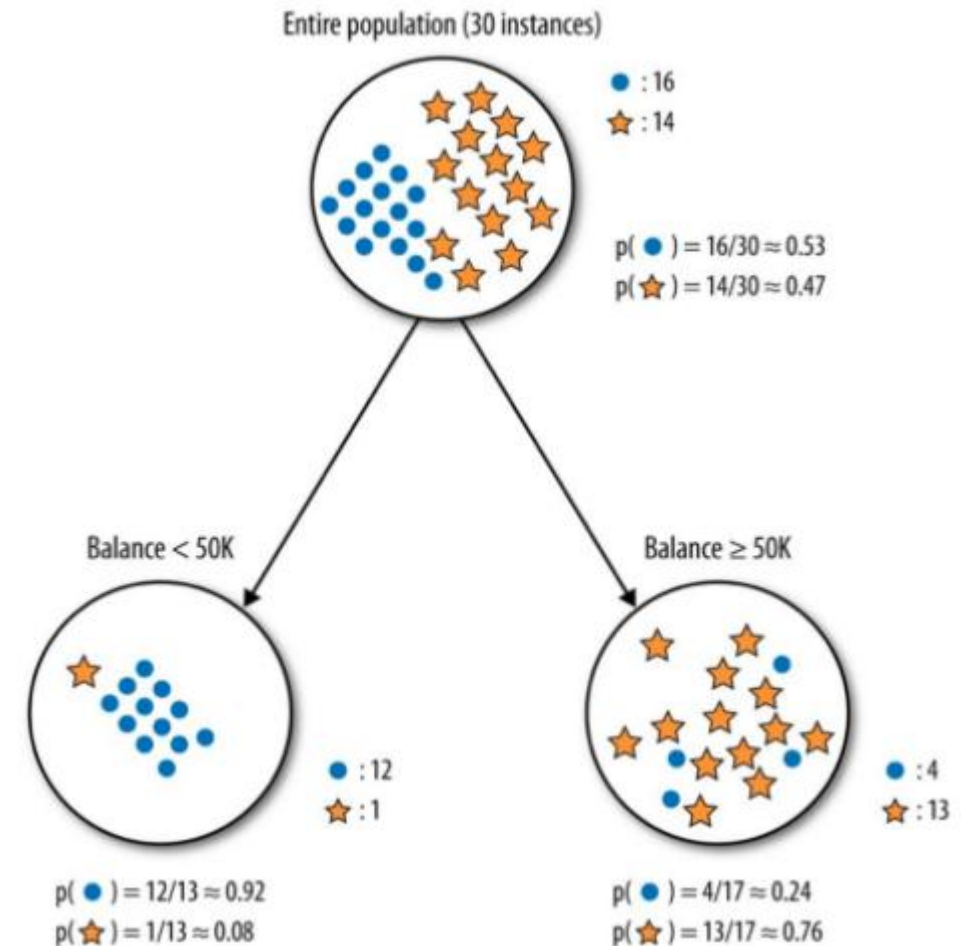
$$= 0.375$$

Example: Decision Tree of Loan

- Create a DT to predict whether a loan given to a person would result in a write-off or not.
- Population: 30 instances
 - 16 belong to the write-off class
 - 14 belong to the non-write-off class
- Two features:
 - “Balance”: “< 50K” or “>50K”
 - “Residence”: “OWN”, “RENT” or “OTHER”.

Example: Decision Tree of Loan

- Classes:
 - Dots: data points with class right-off
 - Stars: the non-write-offs.



Splitting on Balance

- How much uncertainty the tree can reduce by splitting on Balance?

$$E(\text{Parent}) = -\frac{16}{30}\log_2\left(\frac{16}{30}\right) - \frac{14}{30}\log_2\left(\frac{14}{30}\right) \approx 0.99$$

$$E(\text{Balance} < 50K) = -\frac{12}{13}\log_2\left(\frac{12}{13}\right) - \frac{1}{13}\log_2\left(\frac{1}{13}\right) \approx 0.39$$

$$E(\text{Balance} > 50K) = -\frac{4}{17}\log_2\left(\frac{4}{17}\right) - \frac{13}{17}\log_2\left(\frac{13}{17}\right) \approx 0.79$$

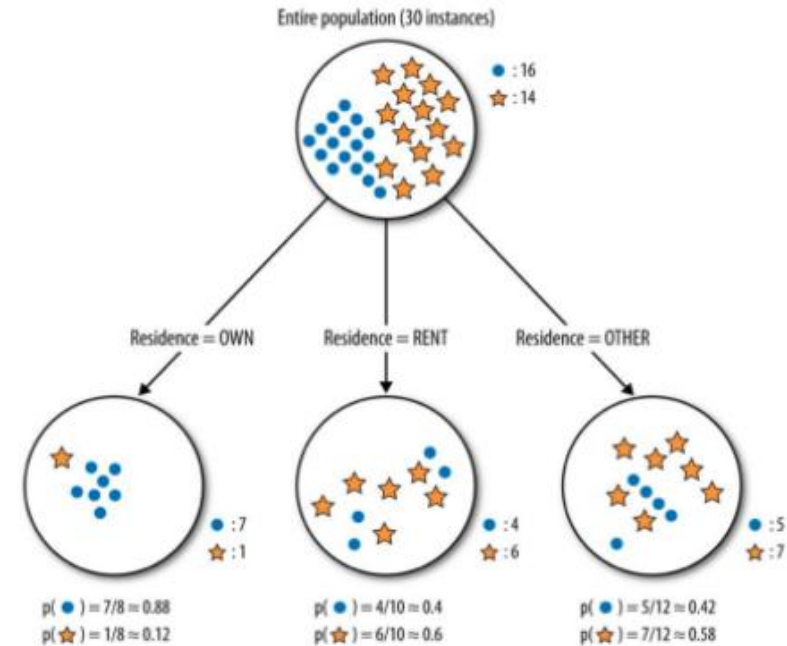
Weighted Average of entropy for each node:

$$\begin{aligned} E(\text{Balance}) &= \frac{13}{30} \times 0.39 + \frac{17}{30} \times 0.79 \\ &= 0.62 \end{aligned}$$

Information Gain:

$$\begin{aligned} IG(\text{Parent}, \text{Balance}) &= E(\text{Parent}) - E(\text{Balance}) \\ &= 0.99 - 0.62 \\ &= 0.37 \end{aligned}$$

Splitting the tree on Residence



Provost, Foster; Fawcett, Tom. Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking

Splitting the tree on Residence

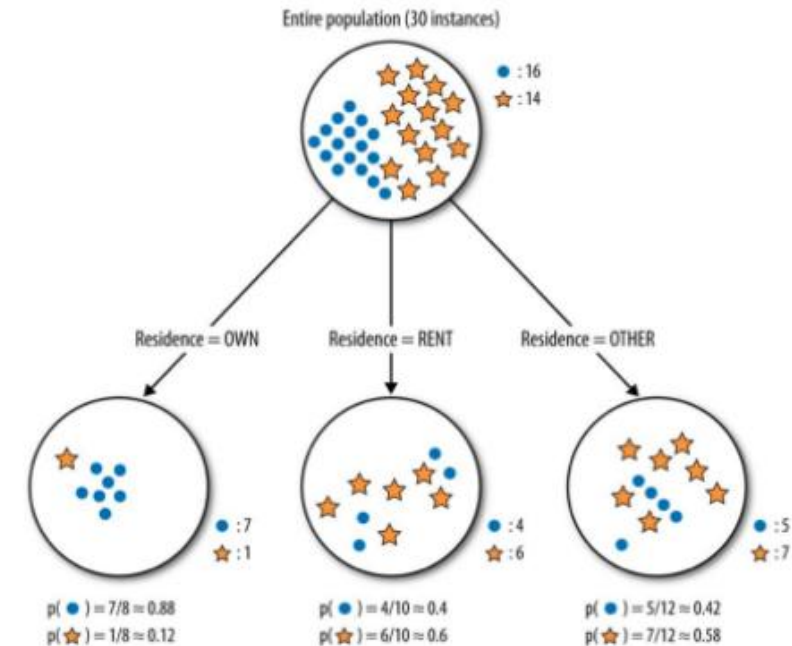
$$E(\text{Residence} = \text{OWN}) = -\frac{7}{8}\log_2\left(\frac{7}{8}\right) - \frac{1}{8}\log_2\left(\frac{1}{8}\right) \approx 0.54$$

$$E(\text{Residence} = \text{RENT}) = -\frac{4}{10}\log_2\left(\frac{4}{10}\right) - \frac{6}{10}\log_2\left(\frac{6}{10}\right) \approx 0.97$$

$$E(\text{Residence} = \text{OTHER}) = -\frac{5}{12}\log_2\left(\frac{5}{12}\right) - \frac{7}{12}\log_2\left(\frac{7}{12}\right) \approx 0.98$$

Weighted Average of entropies for each node:

$$E(\text{Residence}) = \frac{8}{30} \times 0.54 + \frac{10}{30} \times 0.97 + \frac{12}{30} \times 0.98 = 0.86$$



Provost, Foster; Fawcett, Tom. Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking

Information Gain:

$$\begin{aligned} IG(\text{Parent}, \text{Residence}) &= E(\text{Parent}) - E(\text{Residence}) \\ &= 0.99 - 0.86 \\ &= 0.13 \end{aligned}$$

Example: Decision Tree of Loan

- The IG from feature Balance is almost 3 times than the IG from Residence.
- The first split on our data is by using Balance. From here on, this process will be used at every split to identify what feature it is going to split on next.

Ways to Split a Decision Tree

- Continuous Target Variable
 - Reduction in Variance
- Categorical Target Variable
 - Entropy (a measure of randomness)
 - Gini Impurity
 - Information Gain
 - Chi-Square

Splitting Method #1: Reduction in Variance

- A method used when the target variable is continuous, i.e., regression problems.
- It uses variance as a measure for deciding the feature on which node is split into child nodes.

$$Variance = \frac{\sum (X - \mu)^2}{N}$$

Reduction in Variance

The steps to split using reduction in variance:

1. For each split, individually calculate the variance of each child node
2. Calculate the variance of each split as the weighted average variance of child nodes
3. Select the split with the lowest variance
4. Perform steps 1-3 until completely homogeneous nodes are achieved

Splitting Method #2: Information Gain

- For the case of categorical target variable.
- Entropy is used for calculating the randomness or purity of a node.
- **Lower the value of entropy, higher is the purity of the node.**

$$Entropy = - \sum_{i=1}^n p_i \log_2 p_i$$

$$Information\ Gain = 1 - Entropy$$

Splitting Method #3: Gini Impurity

- The target variable is categorical.
- Gini is the probability of correctly labeling a randomly chosen element if it was randomly labeled according to the distribution of labels in the node.

$$\text{Gini Impurity} = 1 - \text{Gini}$$

$$\text{Gini} = \sum_{i=1}^n p_i^2$$

$$\text{Gini Impurity} = 1 - \sum_{i=1}^n p_i^2$$

Gini Impurity

- Gini Impurity is measure of variance across the different classes

$$G(\text{node}) = \sum_{k=1}^c p_k (1 - p_k)$$

Probability of *not* picking a data point from class k

$p_k = \frac{\text{number of observations with class } k}{\text{all observations in node}}$

Probability of picking a data point from class k

Gini Impurity of a node.

Splitting Method #4: Chi-Square

- Used for categorical target values.
- It can create two or more than two splits.
- The statistical significance of differences between the parent node and child nodes.

$$Chi-Square = \sqrt{\frac{(Actual - Expected)^2}{Expected}}$$

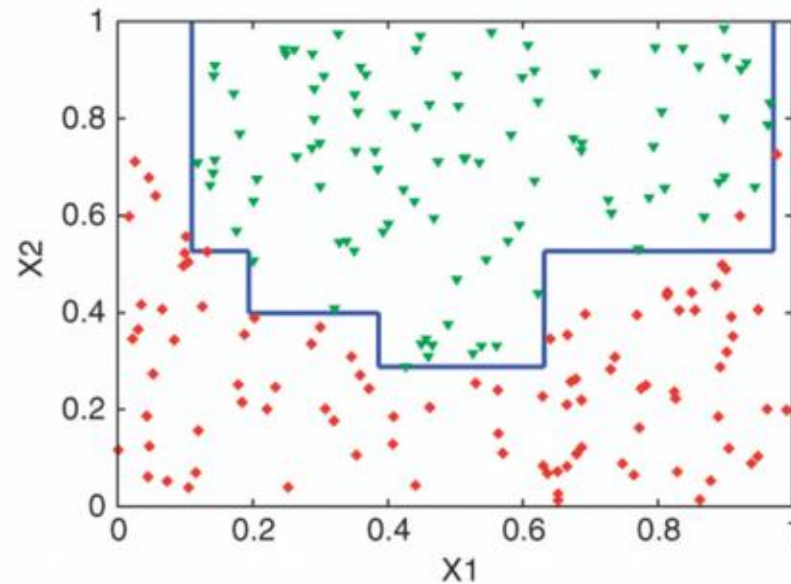
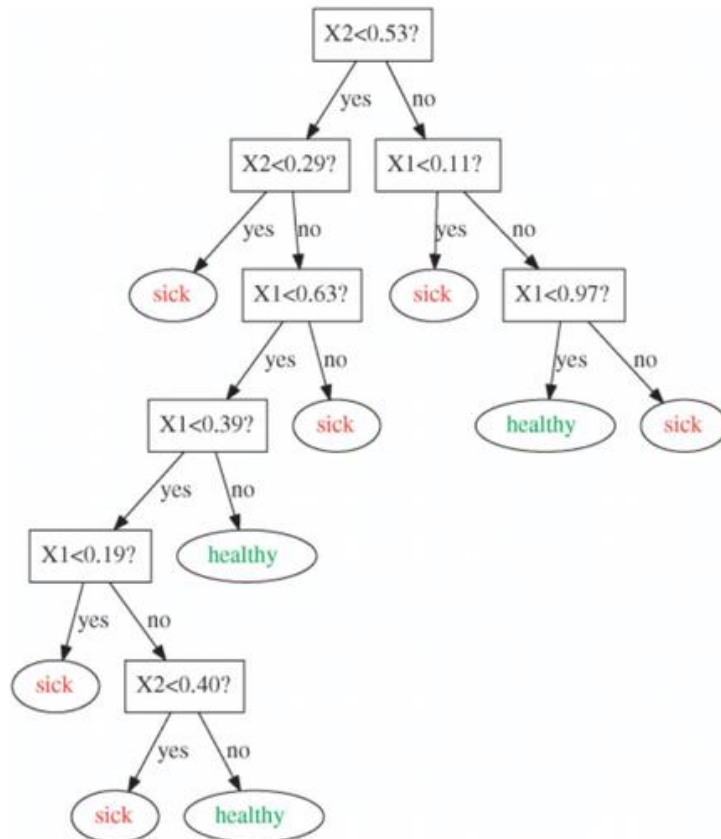
Where, the *Expected* is the expected value for a class in a child node based on the distribution of classes in the parent node, and *Actual* is the actual value for a class in a child node.

Splitting Method #4: Chi-Square

- steps to split a decision tree with Chi-Square:
- For each split, individually calculate the Chi-Square value of each child node by taking the sum of Chi-Square values for each class in a node.
- Calculate the Chi-Square value of each split as the sum of Chi-Square values for all the child nodes.
- Select the split with higher Chi-Square value.
- Until you achieve homogeneous nodes, repeat steps 1-3

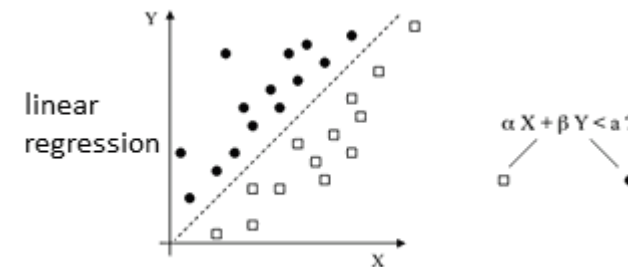
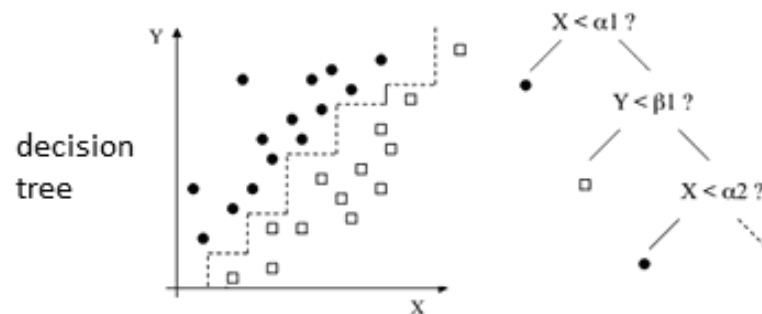
Disadvantage of a Decision Tree

- Tends to overfit data (high variance and false predictions in the test data).



How does DT make decision?

- The decision tree approaches the training data through the sequence of questions rather than understanding the general pattern of the data as linear regression does.



References

- <https://www.analyticsvidhya.com/blog/2020/06/4-ways-split-decision-tree/>
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- <https://towardsdatascience.com/entropy-how-decision-trees-make-decisions-2946b9c18c8>